Adaptive Energy-Aware Navigation for Autonomous Surface Vessels in Complex Ocean Current Fields

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ABSTRACT

In the complex oceanic environment, it is crucial for autonomous surface vessels (ASVs) to navigate along a safe and energy-conserving path. This task is inherently challenging due to the disturbances caused by ocean currents and their complex spatiotemporal variations. Most existing methods consider only the spatial variations measured by sensors but overlook temporal variations. In a time-varying ocean current environment, the addition of the temporal dimension allows for the utilization of currents to reduce the energy consumption of ASVs. In this paper, we introduce a novel approach for efficiently generating energy-conserving paths. By enabling rapid replanning in response to evolving current fields, our method ensures sustainable ASV navigation amid complex spatiotemporal ocean current variations. In detail, we modelled ocean currents within a specific region and integrated these current fields into a randomized path planner, enabling the generation of a path that dynamically navigates the flow field in continuous space. In order to effectively create paths within the present field, we introduce a novel version of the RRT-connect algorithm, named circle-RRT-connect*. The circle-RRT-connect* approach integrates adaptive circle sampling with the traditional RRT* rewiring mechanism. This combination significantly decreases redundant sample points and enhances the connection framework of the RRT tree. By modifying the extension weight and integrating the Euclidean metric relevant to the current field, our method guarantees energy-efficient and collision-free navigation. Through extensive simulations, we thoroughly evaluated our algorithm's performance in complex current environments and conducted comparative analyses with elegant related methods.

1. Introduction

1.1. Background

Autonomous surface vessels (ASVs) are vital tools for ocean exploration and research. These vehicles are widely utilized in a range of applications, including coastal surveillance [1, 2], maritime traffic management [3, 4], search and rescue operations [5], and hydrographic surveying [6]. ASVs are capable of autonomously executing complex marine operations, making them indispensable for advancing our understanding of marine environments and supporting various marine tasks. ASVs navigate to operational areas and complete their missions using autonomous navigation and decision-making systems. In complex marine environments, effective path-planning algorithms are essential for ensuring that ASVs reach their designated areas safely and efficiently.

Path-planning methods for ASVs involve strategically designing trajectories that are optimal or suboptimal while ensuring they remain collision-free. The evaluation of these trajectories typically considers multiple criteria, including trajectory length, temporal duration, and energy expenditure, ensuring that the path planning approach meets the specific operational requirements of the ASV. Given the limited energy supply of ASVs [7, 8], optimizing the path for minimal energy consumption is crucial to maximizing operational endurance and effectiveness. Refining path length while considering dynamic ocean currents and the marine environment allows ASVs to significantly reduce energy consumption without compromising mission success. In marine environments, the pervasive movement of ocean currents can disrupt ASV navigation. However, by strategically utilizing these currents, the ASV's energy consumption can be greatly reduced. Within the domain of ASV path planning, the majority of existing methods [9, 10, 11] aimed at addressing energy consumption tend to focus on isolated factors and lack a comprehensive strategy that considers the various factors influencing energy use.

In the energy-conserving path planning process for ASVs, the key factors influencing energy expenditure include path length and route complexity, velocity and direction of ocean currents, as well as obstacle density and arrangement in the environment. In the paper, we introduce an innovative method that combines obstacle limits, distance measurements, and the dynamics of ocean currents, thereby enabling the efficient formulation of energy-saving navigation routes for ASVs. The traditional RRT-connect method [12] is capable of rapidly generating collision-free paths. However, the resulting path length is often inconsistent and cannot be guaranteed to be as short as possible. In order to generate shorter paths more efficiently, we propose an altered variant of the RRT-connect algorithm, known as circle-RRT-connect*. The process of circle-RRT-connect* involves the combination of adaptive circle sampling and the rewiring technique derived from the conventional RRT* algorithm [13]. This integration significantly reduces unneeded sampling points and improves connectivity within

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the rapid-exploration tree. When applying the circle-RRT-connect* technique in path planning, ocean currents act as an effective vector field that notably accelerates the tree's expansion dynamics. A pivotal advancement in our approach embeds the ocean current's orientation and intensity directly into the expansion mechanism of the circle-RRT-connect* technique. The expansion process can undergo refinement to enhance both energy efficiency and goal-oriented optimization.

Figure 1 presents the simulation results, which clearly demonstrate the efficacy of the proposed method. The detailed framework of this method is outlined in Section 3. The results clearly show the higher efficiency of our approach, which achieves the task using only $482.24J$ energy in 69.83 milliseconds. In contrast, the comparable and elegant method proposed in [11] (detailed in Section 4.1) requires 581.20 milliseconds and consumes substantially more energy, amounting to $574.72J$.

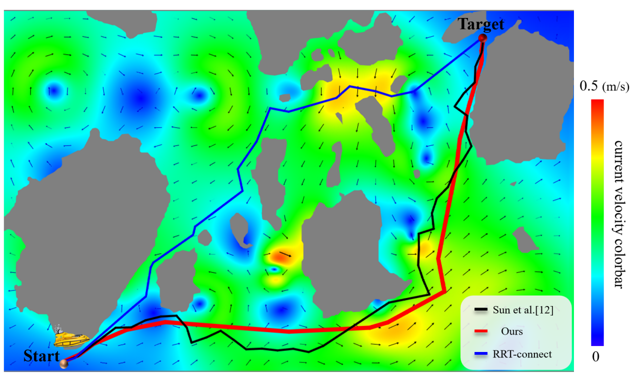


Figure 1: In simulated ocean current environments, our method achieves an energy cost of $482.24J$, completing the task in 69.83 ms with a path length of $1316m$. In comparison, Sun et al.'s algorithm [11] consumes $574.72J$ energy, takes 581.20 ms, and produces a longer path of 1501m. The traditional RRT-connect method, without accounting for ocean currents, generates a shorter path of 1292m but requires $643.11J$ energy and 205.23 ms. The map size is bounded by 1560m $\times$ 1000m.

1.2. Literature review

Numerous scholarly studies have examined path-planning techniques designed specifically for ASVs [4, 14], thereby emphasizing the field's critical importance and central role. This section examines the latest research related to the issue of path planning. The overview is structured into two primary components: (1) path planning incorporating ocean currents and (2) path planning utilizing RRT and its derivations.

1.2.1. Path planning with ocean currents

Within the marine setting where autonomous vehicles are deployed, there are non-uniform ocean currents that dynamically change over time. Therefore, various strategies have emerged to tackle the challenges presented by these ocean currents. Song et al. [15] put forward an innovative multi-layered fast marching approach, which facilitates the creation of ideal trajectories for traversing through continuously changing ocean currents. Ma et al. [16] framed the task of planning paths in the context of ocean currents as a comprehensive nonlinear optimization challenge, encompassing multiple objectives and numerous constraints. Chen et al. [17] developed the OCi-RRT* method, which applies motion constraints to expand the sampling area while factoring in ocean currents during dynamic node speed adjustments. Zeng et al. [18] proposed a path planning strategy that leverages spline curves. Their approach is based on the assumption of a constant propulsion speed, aiming to achieve the shortest possible travel time. Kularatne [19] provided an analytical method for optimizing energy consumption and speed based on ocean current conditions. Soulignac et al. [20] proposed an approach for optimizing energy conservation by selecting an optimal departure speed in time-varying ocean current fields. Kruger et al. [21] enhanced autonomous vehicles path optimization by incorporating time as an additional variable, enabling them to determine the optimal propulsion speed through advanced optimization algorithms. Most methods [11, 22, 23] related to ocean currents in path planning focus solely on isolated factors and lack a comprehensive strategy that considers the various elements influencing energy consumption. In this paper, we are dedicated to integrating multiple factors, including path length, obstacle constraints, and ocean currents, to generate energy-efficient paths for ASVs.

1.2.2. Path planning with RRT and its variants

The rapidly exploring random tree (RRT) method [24] stands as a broadly acknowledged approach, crafted to efficiently address path planning challenges within dynamic and intricate settings. By exploring the configuration space C, RRT incrementally builds a tree structure by extending from randomly sampled points toward previously unexplored regions, allowing efficient identification of viable routes with great effectiveness. However, the algorithm faces several limitations [25], such as low node utilization [26] and unstable paths [27]. To overcome these challenges, a variety of solutions have been proposed. To enhance node utilization, Kalisiak et al. [28] presented the RRT-blossom approach. This approach uses a constraint function for regression to create novel nodes, consequently reducing the probability of investigating redundant areas in the initial search stages. Frazzoli and colleagues [13] introduced the RRT* method. This algorithm attains asymptotic optimality through the integration of random geometric graphs and optimization pruning methods within the node expansion procedure. Nasir et al. [29] unveiled the RRT*-smart approach, blending pioneering sampling techniques with robust path optimization procedures to counter the gradual convergence issues that afflict RRT*. Additionally, multi-directional random trees build upon the single-directional random tree structure by establishing multiple root nodes throughout the workspace. This approach optimizes the connections between these random trees, enhancing overall path planning efficiency. Lavalle et al. [30] introduced the bi-directional RRT method, designed for path planning of high-dimensional robotic arms. This technique requires constructing two distinct trees. One originates from the initial location, while the other arises from the destination. Both trees are methodically expanded until their terminal nodes converge to form a connection. Akgun et al. [31] extended RRT* to develop the asymptotically optimal bidirectional RRT*. In this approach, they first identify an initial path and then apply sampling constraints to minimize the inclusion of invalid nodes, driving the solution towards a localized optimal point. Jordan et al. [32] presented the b-RRT* method incorporating heuristic techniques. Their improvements integrate the rapid convergence properties of RRT-connect [12, 33] with heuristic strategies to alleviate the computational burden associated with node expansion. Despite numerous attempts to overcome the performance constraints of the RRT algorithm, striking an ideal equilibrium between path length and computational efficiency continues to be a significant challenge. In this research, we present the circle-RRT-connect* method as an innovative method for effectively harmonizing conflicting goals, thereby improving the creation of energy-efficient trajectories for ASV navigation.

1.3. Motivation and contribution

This research introduces a method for creating energy-efficient path plans for ASVs, focusing on sustainable navigation in complex ocean current environments. Our research makes the following principal contributions:

- We develop an energy-aware path planning framework in which obstacle constraints, distance measures, and ocean current dynamics are cohesively modelled to optimize autonomous surface vehicle routing.
- We introduce circle-RRT-connect*, a novel RRT-connect variant that boosts path efficiency by coupling adaptive circle-based sampling with a rewiring process, thereby cutting redundant sampling and enhancing connectivity within the tree.
- We undertake a detailed theoretical examination of the circle-RRT-connect* algorithm, revealing a time complexity of $O(n \log n)$ and a space complexity of $O(n)$, where n represents the number of samples. Our analysis indicates that, although the circle-RRT-connect* algorithm demonstrates time performance comparable to that of the basic RRT-connect approach, it provides a unique benefit by achieving a harmonious equilibrium between time efficiency and path optimality.

Our work is organized as follows: In Section 2, we offer background information on dynamic ocean currents and introduce the RRT algorithm. Section 3 outlines our path planning framework tailored for ocean current conditions, with a thorough explanation of the mechanism behind our energy-efficient algorithm. Lastly, we showcase the simulation outcomes and provide analysis in Section 4. For clarity,

Symbols	Description
C	configuration space in oceanic environments
p_s	start point
p_g	goal point
T_s	start tree
T_g	goal tree
D	vortex strength
R	vortex radius
p_{rand}	the random sample node
p_{new}	the newly generated node
p_{near}	a nearest point of p_{rand} adjacent to RRT tree
Π	a feasible path in path planning process
$\mathcal{J}$	a specific cost function.
$\overline{\mathcal{V}_a}$	propulsion speed of ASV
$\overline{\mathcal{V}_s}$	speed relative to the seabed
$\overline{\mathcal{V}_o}$	ocean current velocity
$\mathcal{O}_1, \mathcal{O}_2, \ldots, \mathcal{O}_v$	obstacles in oceanic environments
$O_{p_{new}}$	a sampling circle
$O_{p_{parent}}$	parent circle of $O_{p_{new}}$

Figure 2: Symbols definitions used throughout the following sections.

Figure 2 lists symbols used throughout the following sections.

2. Preliminaries

By fusing active ocean current dynamics with the RRT algorithm and its variants, the technique significantly impacts the development of energy-efficient routes for ASVs operating in complex marine settings. In the subsequent section, we present a detailed analysis of these aspects.

2.1. Dynamic ocean currents

The dynamics of ocean currents hold significant importance in the path planning of ASVs. Neglecting the constraints posed by these currents can result in substantial energy losses, particularly when the ASV is forced to navigate against the flow. In order to enhance energy utilization and to lengthen the operational lifespan of ASVs that rely on finite power reserves, it is crucial to consider ocean currents [34, 35] as a key factor shaping the effectiveness of ASV path planning. Incorporating ocean currents into the planning process enables the preemptive design of ASV routes, allowing for reduced energy losses when navigating against the flow and the potential to harness current-driven thrust to conserve energy. This strategy significantly enhances the feasibility of long-duration missions for ASVs.

Developing an accurate and dependable ocean current model is crucial for investigating the effects of ocean currents on ASV path planning. Ocean currents can exhibit substantial temporal variation due to influences such as wind and tides. Within fluid dynamics, the kinematic aspects governing flow behavior are customarily determined by the Navier-Stokes equations [36]. However, solving the Navier-Stokes equations is highly complex. Consequently, references [37, 38, 39, 40, 41] has introduced a simplified method for characterizing the flow field based on these equations. This approach simulates the ocean current environment by superimposing multiple line vortex models, known as Lamb vortices. It not only maintains a coherent flow structure but also simplifies analysis and processing, making it widely adopted in the path planning research of marine vehicles. The equation for a single Lamb vortex [42] is provided below:

$$\begin{cases} v_x(\vec{c}) = -D\frac{x-x_0}{2\pi||\vec{c}-\vec{c_0}||^2}[1-e^{-(\frac{||\vec{c}-\vec{c_0}||^2}{R^2})}] \\ v_y(\vec{c}) = -D\frac{y-y_0}{2\pi||\vec{c}-\vec{c_0}||^2}[1-e^{-(\frac{||\vec{c}-\vec{c_0}||^2}{R^2})}] \\ ||\vec{c}-\vec{c_0}|| = \sqrt{(x-x_0)^2+(y-y_0)^2} \end{cases} \tag{2.1}$$

Here, v_x and v_y denote the speeds of ocean currents along the x-axis and y-axis, respectively. Additionally, $\vec{c_0}$ marks the location of the vortex center, R represents its radius, and D expresses the vortex's strength. Assume there are n vortices in the map, and after linearly superimposing them, the current velocity at each position $(\mathcal{V}_x, \mathcal{V}_y)$ in the map is represented as:

$$\mathcal{V}_x = \sum_{i=1}^{n} v_x^i, \quad \mathcal{V}_y = \sum_{i=1}^{n} v_y^i \tag{2.2}$$

The occurrence of Lamb vortices in our model is based on several key assumptions. First, we assume an inviscid, incompressible flow regime, where the vorticity is initially concentrated in a single vortex structure, leading to diffusive spreading as described by the single Lamb vortex solution. This form assumes negligible external forcing and relies on the Navier-Stokes equations under high-Reynolds-number conditions, predicting stable vortex evolution over time.

The ASV performs its task over the time period [0, T], during which the ocean current changes at times t_i (where $i = 1, 2, \dots, m$). In practical applications, the larger the value of m, the closer the simulated ocean current is to the real environment. Figure 3 illustrates the ASV's path navigation results amid a marine context defined by three merged vortices based on Eq. 2.1. The black path represents the navigation result without accounting for ocean currents, whereas the red path illustrates the outcome of navigation planning that incorporates ocean current influences. The comparison shows that ocean currents greatly affect the ASV's energy consumption, altering optimal routes and making ocean currents as a critical factor that must be carefully considered in ASV path planning.

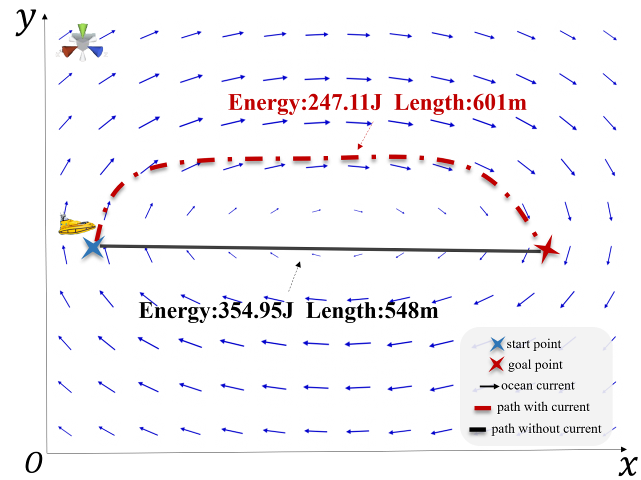


Figure 3: Path navigation results were obtained under conditions involving three overlapping vortices. The resulting trajectory (displayed in red), which incorporates ocean current influences, operates using only 347.11J energy while exhibiting a path length of 601m. Conversely, the alternative route (shown in black), disregarding ocean effects, consumes 354.95J energy and has a path length of 548m.The current speed ranges from 0 to 0.5 m/s and the map size is bounded by 600m × 500m.

2.2. Rapidly-exploring random tree (RRT)

When navigating autonomous vehicles, the RRT algorithm [24, 26, 43] offers significant advantages, as it efficiently explores intricate spaces. RRT demonstrates excellence in sequentially generating search trees from the initial position, allowing it to navigate around obstacles and adapt to dynamic environments. Its random sampling approach facilitates flexible and feasible pathfinding even in partially unknown or changing environments. In detail, RRT starts with an initial tree T rooted at the start point p_s. During every iteration, the random sample p_{rand} gets generated inside the configuration space C, and the nearest neighbor p_{near} in T is identified. A new node $q_{new} = p_{nearest} + \eta \cdot \frac{p_{rand}-p_{near}}{\|p_{rand}-p_{near}\|}$, where η is the step size, is created by moving from p_{near} towards p_{rand}. The edge (p_{near}, p_{new}) is added to T if there is no collision along the path connecting them. The process repeats until T reaches the goal point p_{goal} or a stopping criterion $\|p_{new} - p_g\| \leq \epsilon$ is met. This procedure is illustrated in Figure 4(a). The inherent randomness in RRT algorithm offers substantial advantages by enabling adaptability to diverse and dynamically evolving environments. However, this randomness may also lead to the generation of suboptimal paths that are longer or less efficient. Furthermore, utilizing a constant step size in the fundamental RRT may lead to excessively rough incremental advancements, which impedes the algorithm's ability to explore the search space efficiently and can extend the time required to identify a viable path. In order to overcome these constraints, several modifications of the RRT algorithm have been suggested, with notable examples including RRT-connect [12] and RRT* [13].

The RRT-connect algorithm operates by incrementally building two trees $T_s = (V_s, E_s)$ and $T_g = (V_g, E_g)$, rooted at the starting point $p_s \in C$ and target point $p_g \in C$ illustrated in Figure 4(b). At each iteration, the algorithm samples a random point $p_{\text{rand}} \in C$, and identifies the closest point

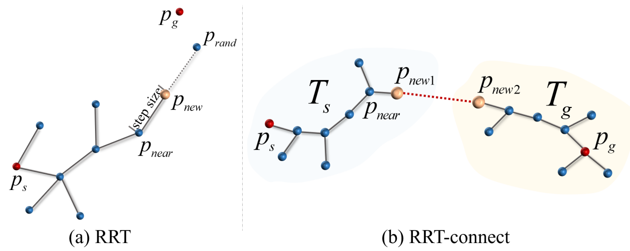


Figure 4: The visualization of tree formations in the RRT and RRT-connect methods. (a) By randomly sampling points and stretching branches towards unexplored areas, the RRT method progressively enlarges a tree. (b) The RRT-connect method employs two trees, T_s and T_g, to efficiently explore the configuration space C, ultimately converging towards one another.

$p_{\text{near}} \in V_{\text{s}}$ in start tree. It attempts to extend from p_{near} towards p_{rand} by a fixed step size η, yielding a new point p_{new}. The algorithm terminates when p_{new} in one tree (e.g., T_s) is within a distance ϵ of the nearest node in the other tree (e.g., T_g), i.e., $\|p_{\text{new}} - p_{\text{nearest}}\| < \epsilon$, at which point is attempted to link the trees and form a path. The RRT-connect method boosts the basic RRT algorithm's convergence speed by utilizing a two-way search strategy. This reduces the time to find a valid path, as the trees are more likely to meet in the middle rather than relying on one tree to explore the entire space. However, RRT-connect method is effective for fast path planning, while it does not guarantee that the final trajectory represents the shortest or most energy-efficient option.

The RRT* algorithm extends the basic RRT by incrementally building a single tree $T = (V, E)$, rooted at the starting point $p_{\text{s}} \in C$, while aiming for asymptotic optimality towards the target point $p_{\text{g}} \in C$. At each iteration, the algorithm samples a random point $p_{\text{rand}} \in C$ and identifies the closest point $p_{\text{near}} \in V$. It attempts to extend from p_{near} towards p_{rand} by a fixed step size η, yielding a new point p_{new}, provided the extension is collision-free. Unlike standard RRT, RRT* incorporates a rewiring step: it selects the parent for p_{new} from a set of nearby nodes that minimizes the path cost (e.g., Euclidean distance or other metrics) from p_{s} to p_{new}. Additionally, it checks neighboring nodes in a radius around p_{new} and rewires them to p_{new} if doing so reduces their total cost from the root. This process ensures that the tree converges to an optimal path as the number of samples increases. While RRT* improves upon RRT by providing probabilistic guarantees of optimality, it requires more computational effort per iteration due to the nearest-neighbor searches and rewiring operations, making it suitable for scenarios where path quality is prioritized over rapid initial solutions.

Improving both efficiency and path quality holds significance in path-planning methods for advancing research and practical applications. Thus, the research community relentlessly strives to discover an optimally efficient routing solution that upholds superior temporal performance. To satisfy the dual requirements for quick temporal efficiency and robust path quality in navigation, we introduce an enhanced variant of the RRT-connect method, designated as circle-RRT-connect*. By merging the rapid exploratory power of RRT-connect with the asymptotical optimality characteristic of RRT*, our method substantially improves the algorithm's overall performance. For an in-depth technical explanation, kindly refer to Section 3.2.

3. Path planning framework in ocean current environments

This paper investigates an energy-saving route planning strategy for ASVs functioning in marine environments, taking into account ocean currents and intricate obstacles while ensuring collision avoidance. The available environment space is partitioned into a grid pattern and subsequently transformed into a continuous domain. The following assumptions serve as the foundation for the algorithm: (1) The ocean currents, obstacles, and the ASV are on the same horizontal plane. (2) The ASV can obtain environmental information, such as ocean current speed and obstacle positions, using sensors like current meters and sonar systems. This paper utilizes the Micron DST, a compact sonar model for digital imaging, which was developed by Tritech, for creating maps and detecting environmental information [44].

3.1. Fundamental problem statement

Let the configuration space C is assumed to be a bounded subset of $\mathbb{R}^2$, and $C_o \subset C$ represent the region occupied by obstacles, which is assumed to be closed and bounded. The set $C_f := C \setminus C_o$ is referred to as the obstacle-free space. The points marking the beginning and end of the path planning are p_s and p_g, respectively. The task of discovering a collision-free trajectory is defined as the path-planning problem (C, p_s, p_g). A collision-free trajectory is described by a continuous function $\Pi : [0, 1] \to C$, such that $\Pi(\tau) \in C_f$ for all $\tau \in [0, 1]$, with $\Pi(0) = p_s \in C_s$ and $\Pi(1) = p_g \in C_g$, where C_s and C_g signify the starting and destination regions, respectively.

Definition 3.1. *(**Feasible path planning.**) In the considered path planning problem (C, p_s, p_g), The aim is to discover a viable route Π such that Π is collision-free, $\Pi(0) = p_s$, and $\Pi(1) = p_g$.*

Definition 3.2. *(**Optimal path planning.**) Considering a defined path planning problem (C, p_s, p_g), the goal is to determine the most optimal path Π^* that minimizes the cost function $\mathcal{J}(\Pi)$. The path Π^* must satisfy the condition $\mathcal{J}(\Pi^*) = \min\{\mathcal{J}(\Pi) : \Pi \text{ is a valid path}\}$, where $\mathcal{J}(\Pi)$ signifies the expense incurred to attain the desired goal state p_g along the path Π.*

In addition, the task of path planning is defined as reducing the search required to identify the optimal trajectory Π^*

based on a specified cost function $\mathcal{J}$, as detailed below:

$$\begin{aligned} &\textbf{Minimize} \quad \mathcal{J}(\Pi_i) \\ &\textbf{s. t.} \quad \Pi(0) = p_s,\ \Pi(1) = p_g, \\ &\quad \{i = 1, 2, ..., k\}. \end{aligned} \tag{3.1}$$

Π_i represents one path among several possible paths connecting the starting point p_s and the endpoint p_g. Subject to the constraints defined in Eq. 3.1, we aim to identify an energy-efficient path within complex ocean current environments efficiently.

3.1.1. Path structure

The path Π consists of an ordered sequence of waypoints, denoted as $(p_s, p_1, p_2, \ldots, p_k, p_g)$, where each consecutive pair of waypoints is connected by a straight line segment. These waypoints can reside at any location within the free configuration space C_f, and the path length is mathematically represented as follows:

$$L(\Pi) = \sum_{i=0}^{k} \|\overrightarrow{p_i p_{i+1}}\| \tag{3.2}$$

Here, $p_0 = p_s$ and $p_{k+1} = p_g$. This formulation addresses the limitations of conventional grid-based path planners by overcoming the discrete motion direction constraints. Based on Eq. 3.1 and the given cost function $\mathcal{J}$, Eq. 3.2 can be further extended to define the weighted path length L_w as follows:

$$L_w(\Pi) = \sum_{i=0}^{k} \mathcal{J}(\overrightarrow{p_i p_{i+1}}) \|\overrightarrow{p_i p_{i+1}}\| \tag{3.3}$$

In ASV navigation, the cost function $\mathcal{J}$ provides a comprehensive representation of environmental factors, including wind, ocean currents, and salinity, that influence the vehicle's trajectory. In this study, we focus specifically on ocean currents as a critical factor, integrating them into the navigation process to significantly enhance energy efficiency as mentioned in Section 3.1.2.

3.1.2. Energy consumption in path planning

The aggregate energy E_{total} consumed by the ASV along the path Π is determined by summing the energy expended over each sub-path segment between successive points p_i and p_{i+1}, where $i = 0, 1, \ldots, k$, as defined below:

$$E_{total} = \sum_{i=0}^{k} E(\overrightarrow{p_i p_{i+1}}) \tag{3.4}$$

$E(\overrightarrow{p_i p_{i+1}})$ denotes the consumed energy of the sub-path $\overrightarrow{p_i p_{i+1}}$. From a mechanics standpoint, the ASV's energy usage over the sub-path $\overrightarrow{p_i p_{i+1}}$ is given by $P\Delta t$, where P represents the propulsion power of the ASV, which is directly correlates with the cube of the propulsion speed of the ASV [45], determined by the ASV's design. Consequently, the total energy E_{total} of the ASV can be further defined as:

$$E_{total} = \sum_{i=0}^{k} c\|\mathcal{V}_a(\overrightarrow{p_i p_{i+1}})\|^3 \cdot \frac{\|\overrightarrow{p_i p_{i+1}}\|}{\|\mathcal{V}_s(\overrightarrow{p_i p_{i+1}})\|} \tag{3.5}$$

Here, $\mathcal{V}_a$ and $\mathcal{V}_s$ denote the propulsion speed of the Autonomous Surface Vehicle (ASV) and its speed relative to the seabed, respectively, while traversing the designated sub-path $\overrightarrow{p_i p_{i+1}}$. The drag coefficient is represented by c, whereas $\mathcal{V}_a$ is a constant value determined by the ASV's design. The speed relative to the seabed, denoted as $\mathcal{V}_s$, is obtained through the vector summation of the ASV's propulsion speed $\mathcal{V}_a$ and the marine current speed $\mathcal{V}_o$, as given by the following equation:

$$\overrightarrow{\mathcal{V}_s} = \overrightarrow{\mathcal{V}_a} + \overrightarrow{\mathcal{V}_o} \tag{3.6}$$

Specifically, for a path segment $\overrightarrow{p_i p_{i+1}}$ from point p_i to p_{i+1}, with initial current velocity $\mathcal{V}_o(p_i)$ at p_i and terminal current velocity $\mathcal{V}_o(p_{i+1})$ at p_{i+1}, the average current velocity $\mathcal{V}_o(\overrightarrow{p_i p_{i+1}})$ used in Equation 3.5 is calculated as $\mathcal{V}_o(\overrightarrow{p_i p_{i+1}}) = \frac{\mathcal{V}_o(p_i)+\mathcal{V}_o(p_{i+1})}{2}$.

As indicated in Eq. 3.5, the ASV's energy consumption is closely tied to its trajectory Π and relative speed $\mathcal{V}_s$. During the ASV's navigation, to achieve an energy-optimal path while maintaining the desired path length, selecting an appropriate $\mathcal{V}_s$ based on the ocean current is essential. The proposed method in this paper extends classical RRT-based path planning algorithms by integrating the ASV's relative speed $\mathcal{V}_s$ as a critical search dimension. The formulation of the path $\Pi(p_s, p_g)$ is expressed as follows:

$$\Pi(p_s, p_g) = \{(\overrightarrow{p_s p_1}, \mathcal{V}_s^1), \ldots, (\overrightarrow{p_i p_{i+1}}, \mathcal{V}_s^i), \ldots, (\overrightarrow{p_k p_g}, \mathcal{V}_s^k)\} \tag{3.7}$$

$\mathcal{V}_s^i$ denotes the relative speed of the ASV in the i-th path segment. The optimal energy consumption path planning model under an ocean current field is formulated as follows:

$$\begin{aligned} &\textbf{Minimize} \ \ E_{total}(\Pi) = \sum_{i=0}^{k} c\|\mathcal{V}_a(\overrightarrow{p_i p_{i+1}})\|^3 \cdot \frac{\|\overrightarrow{p_i p_{i+1}}\|}{\|\mathcal{V}_s(\overrightarrow{p_i p_{i+1}})\|} \\ &\textbf{s. t.} \quad t \in [0, T],\ \mathcal{V}_{min} \le \mathcal{V}_a(\overrightarrow{p_i p_{i+1}}) \le \mathcal{V}_{max} \end{aligned} \tag{3.8}$$

Since ocean currents vary over time, the constraint $t \in [0, T]$ limits the ASV's departure time to the interval $[0, T]$. The parameters $\mathcal{V}_{min}$ and $\mathcal{V}_{max}$ establish bounds on the ASV's propulsion speed $\mathcal{V}_a$, ensuring that the vehicle operates within a predefined speed range during actual navigation. Note that the relative speed $\mathcal{V}_s$ is determined by the vector summation of $\mathcal{V}_a$ and the marine current speed $\mathcal{V}_o$, as expressed in Eq. 3.6.

In summary, as guided by Eq. 3.8, to enable ASV navigation that is both energy-efficient and collision-avoidant, we propose an enhanced variant of the RRT-connect approach, which we call circle-RRT-connect*. This approach prioritizes the optimization of path length while also taking into account the impact of ocean currents. We modelled ocean currents within a designated region and incorporated them into the circle-RRT-connect* algorithm. This integration enables the utilization of current-induced thrust to optimize energy efficiency while simultaneously striving to minimize the path length. For further technical specifics, please refer to Section 3.2.

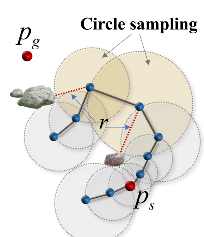
3.2. Circle-RRT-connect*

In ASV path navigation, under identical environmental conditions, shorter navigation paths clearly lead to reduced energy consumption during task execution. we propose circle-RRT-connect*, which combines RRT*'s asymptotic optimality with RRT-connect's efficiency. The technical details are as follows.

3.2.1. *Key technical points*

Building on the above discussion, we highlight the key technical aspects of the circle-RRT-connect* algorithm, including circle sampling, energy-based path segments, parent node selection, and the rewiring operation. Each of these components has a significant advancement from traditional RRT and its variants, as detailed in the following paragraph.

Circle sampling. In the circle-RRT-connect* algorithm, circle sampling serves as the fundamental sampling mechanism. Unlike traditional RRT-connect, where nodes are represented by points, here the nodes correspond to circles with defined areas.



Each circle is associated with a disk of radius r, and the sampled circles are positioned at specified distances from one another to ensure proper spacing. As shown in the inset on the right, it is not possible to position any new sampled circles inside the region already occupied by an existing circle. Within the enhanced RRT-connect method, the **empty disk** characteristic plays an important role by making certain that fresh samples are positioned at safe intervals throughout the tree framework. By maintaining this condition, the algorithm reliably covers open areas throughout its search, which in turn increases its effectiveness at discovering new, previously unexplored spaces. With a set number of circle samples, this property guarantees an optimal sample distribution across the entire configuration space. By adopting this strategy, the prevalent issue of producing redundant points that frequently emerge during the conventional random sampling procedure employed by the classical RRT-connect algorithm is efficiently addressed.

Energy-based path segments. In ocean environments, current conditions can vary considerably both across different regions at the same time and within the same region at different times. The consumed energy of ASVs during navigation is significantly impacted by these variations in ocean currents. Figure 5 illustrates the variation in energy consumption for ASVs navigating the same region at different times. Consequently, incorporating ocean current data is crucial for achieving energy-efficient path planning, especially under complex and dynamic conditions. In the proposed circle-RRT-connect* algorithm, the feasible path Π_f generated by the RRT tree T consists of a sequence of path points $(p_s, p_1, \dots, p_k, p_g)$. The consumed energy of the ASV along Π_f is calculated as the sum of the energy consumed along each sub-path $\Pi_{(i:i+1)} = \overrightarrow{p_i p_{i+1}}$ where $i = 0, 1, \dots, k$. Minimizing the energy consumption of each sub-path, while

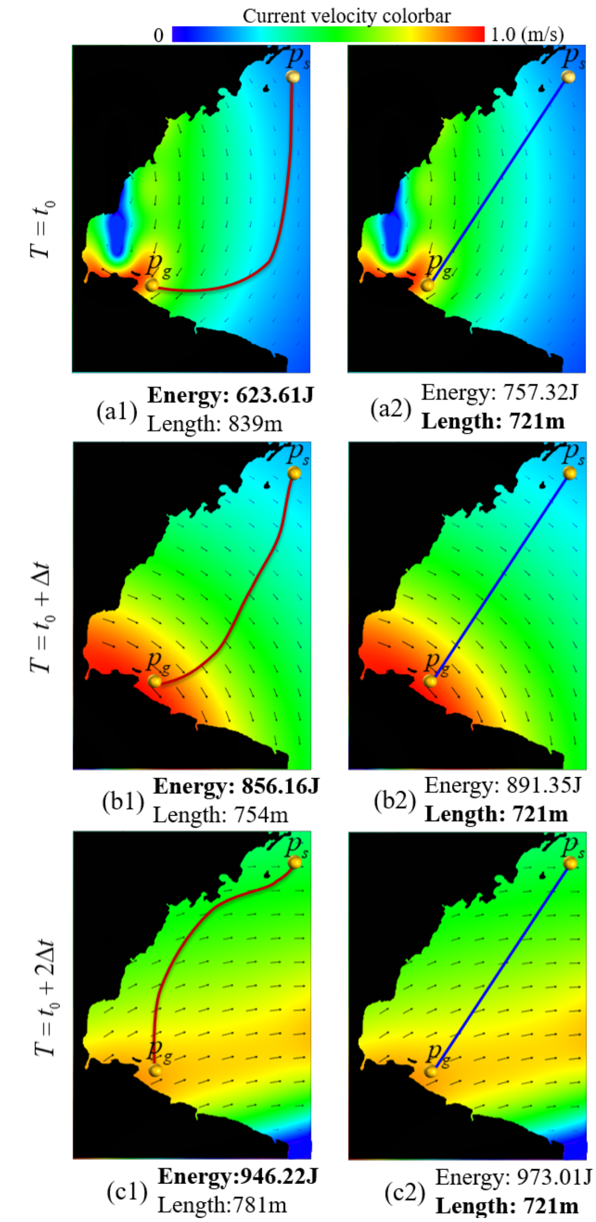


Figure 5: Under time-varying ocean current conditions, we compare the outcomes of energy-efficient and length-optimal navigation strategies over consecutive intervals t_0, $t_0 + \Delta t$, and $t_0{+}2\Delta t$, where Δt denotes the time interval. (a1)~(c1) Energy-saving navigation results; (a2)~(c2) Length-optimal navigation results.

ensuring path feasibility, guarantees optimal energy use for the entire route. As expressed in Eq. 3.5, the energy consumption $E(\Pi_{(i:i+1)})$ for each sub-path $\Pi_{(i:i+1)}$ can be formulated as follows:

$$E(\Pi_{(i:i+1)}) = \frac{c\|\mathcal{V}_a(\Pi_{(i:i+1)})\|^3 \cdot \|\Pi_{(i:i+1)}\|}{\|\overrightarrow{\mathcal{V}_a}(\Pi_{(i:i+1)}) + \overrightarrow{\mathcal{V}_o}(\Pi_{(i:i+1)})\|} \tag{3.9}$$

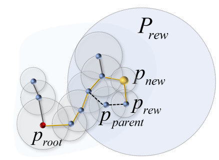
where $\overrightarrow{\mathcal{V}_a}$ denotes the ASV's velocity vector and $\overrightarrow{\mathcal{V}_o}$ represents the ocean current velocity vector. Therefore, assuming a constant ASV speed $\overrightarrow{\mathcal{V}_a}$, the energy consumption along each sub-path is governed by two primary factors: (1) The length of the sub-path and (2) The ocean current velocity $\overrightarrow{\mathcal{V}_o}$. To achieve optimal energy efficiency, the ASV should navigate in alignment with the ocean current direction whenever possible. The algorithm introduced in this study accounts for both factors, optimizing the energy consumption along each sub-path.

Parent nodes selection. In the traditional RRT* algorithm, path quality is improved by evaluating the newly added node p_{new} and its neighboring nodes. The algorithm calculates the cost of paths from p_{new} through these neighboring nodes to the tree root p_{root}. Afterward, the node with the lowest overall cost is chosen to serve as the parent for p_{new}, thereby steering the path closer to the optimal solution. Building upon this concept, our method extends it by identifying the lowest-energy parent node p_{parent} within the candidate sampling set P_{near}. Notably, by leveraging a sampling range that covers the same set of nodes, the circle-RRT-connect* algorithm enables a broader exploration space for selecting p_{parent}. This expanded search space increases the likelihood of discovering shorter paths, thus enhancing overall path optimality. As discussed previously, optimizing energy efficiency during the ASV's navigation requires aligning its movement direction with the prevailing ocean currents as closely as possible. This alignment enhances the ASV's forward motion while minimizing energy consumption. Therefore, during the parent node selection process, unlike the traditional RRT*, which bases parent node selection only on the length of the path, the circle-RRT-connect* method instead identifies the candidate within the set P_{near} possessing the lowest energy cost and designates it as the target parent node p_{parent}. Let $E(\Pi(p_{root}, p_{new}))$ denote the energy consumption of the path $\Pi(p_{root}, p_{new})$ from the circle tree root p_{root} to the new node p_{new} via a candidate parent node p_{cad} in the neighboring set P_{near}. The algorithm then identifies the node with the lowest aggregate cost and assigns it as the parent node p_{parent} of p_{new}, as described below:

$$\begin{aligned} \textbf{Minimize} \quad & E(\Pi(p_{root}, p_{cad\_i})) + E(\Pi(p_{cad\_i}, p_{new})) \\ \textbf{s. t.} \quad & p_{cad\_i} \in P_{near},\ i = \{1, 2, ..., m\}. \end{aligned} \tag{3.10}$$

This approach prioritizes energy efficiency over path length, aligning with the optimization goals for ASV navigation.

$$\begin{aligned} &\big(E(\Pi(p_{root}, p_{new})) \ + \ E(\Pi(p_{new}, p_{rew}))\big) \\ &\qquad\qquad < \\ &\big(E(\Pi(p_{root}, p_{parent})) \ + \ E(\Pi(p_{parent}, p_{rew}))\big) \end{aligned} \tag{3.11}$$



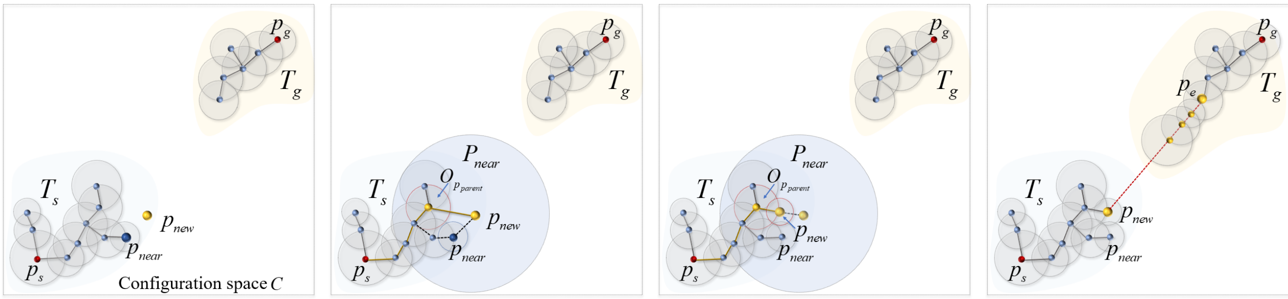
Rewiring operation. Within the RRT* method, performing the rewiring step plays an essential role, as it improves the stability of the solution by reducing the total length of the path. Building on the asymptotic optimality principles established by RRT*, our goal is to incorporate this feature into the circle-RRT-connect* method. In contrast to conventional methods that rely only on path length to determine routes, our approach integrates ocean current information, enabling the planning of paths that are more energy-efficient. Let p_{new} denote the new node extended in the circle tree while the subset P_{rew} consists of candidate node points that need to be optimized, as depicted in the upper inset. If the connection between p_{new} and p_{rew} of the subset P_{rew} remains collision-free, while the local path Π from the root p_{root} to p_{rew} through p_{new} exhibits a reduced energy expenditure compared to the current route as defined in Eq. 3.11, then the parent node p_{parent} of p_{rew} is replaced with p_{new}.

3.2.2. Algorithm framework and details

In the configuration space C, let P represent the point set, S denote the collection of circles, and $\mathbb{E}$ be the collection of edges. The initialization involves constructing two circle trees T_s and T_g, rooted at the start point p_s and the goal point p_g. Upon expanding the fundamental RRT-connect structure, we present the circle-RRT-connect* method. The process flow is depicted in Fig. 6, and the associated pseudocode is outlined in Algorithm 1. A comprehensive description of the primary steps employed in the circle-RRT-connect* approach is provided below, further clarifying its procedures.

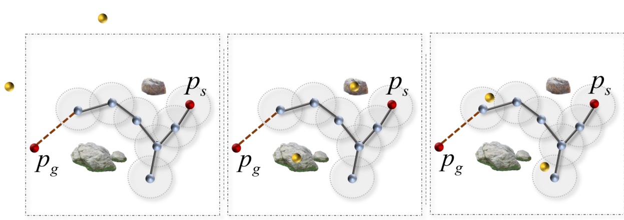
#Step* 1 *Sampling *(Line 3)*. Following the initialization of the trees T_s and T_g with the point set P, circle set S, and edge set $\mathbb{E}$ (Line 1), a random point is sampled within the configuration space C. It is essential that the sampling point p_{new} is positioned at the center of the circle sample. Three distinct scenarios arise in which sampling is strictly prohibited as illustrated in Fig. 7.

#Step* 2 *Lowest energy node confirmation *(Line 10)*. After acquiring the sampling point p_{new}, the algorithm begins an extension process to grow the path-planning tree. After obtaining p_{new}, the circle-RRT-connect* approach locates the nearby node p_{near} in the tree that results in the least energy usage, as calculated according to Eq.3.9. The identification of the node with the lowest energy consumption is pivotal for optimizing the algorithm's overall performance. To enhance search efficiency, we modify an algorithm [46] that employs locality-sensitive hashing for the efficient clustering of proximate points. (Figure 6(a)). The improved algorithm gives identical hash values to neighboring nodes,



(a) Sampling & Lowest energy node confirmation (b) Choose circle parent (c) Circle tree expansion (d) Bi-direction expansion

Figure 6: Pipeline of circle-RRT-connect*. (a) A point p_{new} is randomly sampled within the free configuration space C and the lowest-energy node p_{near} is identified based on Eq. 3.9. (b) The circle parent node $O_{p_{parent}}$ of p_{new}, which incurs a lower energy cost, is identified from the subset P_{near}, replacing the candidate parent node p_{near} of p_{new}. (c) Point p_{new} is projected onto the edge of the parent circle $O_{p_{parent}}$ and utilized as the centre of the sampling circle. (d) The energy-efficient node p_e is identified in the opposing tree T_g and a series of intermediate nodes is generated along the straight line $\overrightarrow{p_e p_{new}}$.



(a) (b) (c)

Figure 7: Instances of flawed samples (highlighted in yellow). (a) The samples reside beyond the configuration space C; (b) The samples are situated in obstacles; (c) The samples are located within previously established circle samples.

grouping them together within the same bucket of the hash table according to these shared values. Utilizing this data structure, our approach allows for rapidly finding the neighbor with minimal energy use for any specific point, typically in near-constant time, since it only involves accessing the appropriate bucket within the hash table.

***#Step* 3 *Choose circle parent* $O_{p_{parent}}$** *(Line 12-14)*. In the **ChooseCircleParent** procedure of circle-RRT-connect*, a meticulous process is employed to derive a subset P_{near} from the existing point set P. The subset denoted as P_{near} is comprised within a hypersphere having a radius of $\mathcal{R}$ and centered around p_{new}. In the framework, the process of selection seeks to identify the node p_{parent} from P_{near} to replace the candidate parent node p_{near} of p_{new} (Figure 6(b)). This replacement endeavor is focused on decreasing the energy expenditure of the path from the starting point p_s to p_{new}, as indicated by Eq. 3.10. The goal is to identify the relay point p_{cad} within P_{near} that exhibits the lowest energy consumption, thereby serving as the parent point for p_{new}. Once p_{cad} is confirmed as the parent point of p_{new}, we can designate the corresponding circle in the circle set S as the parent circle node $O_{p_{parent}}$ of p_{new}.

#Step* 4 *Circle tree expansion *(Line 16-23)*. After completing the earlier procedures, the created random point p_{new} is projected onto the boundary of parent node $O_{p_{parent}}$. During the culminating stage of the circle sampling procedure, p_{new} serves as the centre for the sampling circle $O_{p_{new}}$. The radius r of $O_{p_{new}}$ is ascertained by computing the distance between p_{new} and the nearest obstacle point within $\{\mathcal{O}_1, \mathcal{O}_2, ..., \mathcal{O}_v\}$ in the configuration space C (Figure 6(c)). At last, the valid sampling circle $O_{p_{\text{new}}}$ is integrated in the tree T_s, while simultaneously, the edge connecting p_{parent} to p_{new} is added to the set $\mathbb{E}$.

#Step* 5 *Bi-direction expansion *(Line 25)*. After the existing tree T_s extends to reach a fresh node p_{new}, taking inspiration from the dual-direction growth found in the original RRT-connect technique, the proposed approach searches for the node p_{e} with optimal energy efficiency within the opposite tree T_{g} (see Figure 6(d)), such that the following condition is satisfied:

$$E(\Pi(p_{new}, p_e)) = \arg\min_{\widetilde{p_e} \in T_{\text{g}}} E(\Pi(p_{new}, \widetilde{p_e})).$$

The algorithm then generates a series of intermediate nodes along the straight line from p_{e} to p_{new}, with each intermediate node p_{mid}. Each p_{mid} undergoes collision checking **CollisionCheck** procedure to ensure no obstacles. If the connection is successful, the trees merge to form a complete path. In the event of a collision or a failed connection, the trees are swapped, and the process persists until either a path is discovered or the preset iteration limit is attained.

3.3. Theoretical analysis

In this part, our aim is to perform an in-depth evaluation in order to thoroughly demonstrate the theoretical optimality and examine the computational efficiency of the circle-RRT-connect* algorithm.

Property 3.1. *When the count of samples, denoted by n, approaches infinity ($n \to \infty$), the likelihood of circle-RRT-connect* tending towards the optimal solution for the specified path-planning challenge can be stated as follows:*

$$\mathbb{P}\left(\lim_{n\to\infty} L(\Pi) = L(\Pi^*)\right) = 1.$$

Algorithm 1 Circle-RRT-connect*

Input: 1.Configuration space $\mathcal{C}$, start and goal point p_s, p_g;
2.Point set P, circle set S and edge set $\mathbb{E}$.
Output: An energy-efficient path Π from p_s to p_g.
1: T_s.**init()**, T_g.**init()**.
2: **while NoConnect**(T_s,T_g) **do**
3: p_{new} = **Sample**($\mathcal{C}$);
4: **if** p_{new} is valid **then**
5: Goto step 10;
6: **else**
7: Goto step 3;
8: **end if**
9: /* Locate lowest energy node based on Eq.3.9. */
10: p_{near} = **NearLowestEnergyNode**(T_s, p_{new});
11: **if CollisionCheck**(p_{near}, p_{new}) **then**
12: P_{near} = **CandidateParentSet**(P, p_{new}, $\mathcal{R}$);
13: /*Find energy-efficient parent by Eq.3.10. */
14: $O_{p_{parent}}$ = **ChooseCircleParent**(P_{near}, p_{new}, p_{near});
15: /* Project p_{new} onto boundary of $O_{p_{parent}}$. */
16: p_{new} = **PointProject**(p_{new},$O_{p_{parent}}$);
17: /* $\mathcal{D}_s$ is the shortest distance from p_{new} to obstacles $\{\mathcal{O}_1, \mathcal{O}_2 \dots \mathcal{O}_v\}$. */
18: $O_{p_{new}}$ = **CircleSampling**(p_{new}, $\mathcal{D}_s$);
19: /* Circle tree T_s expansion. */
20: T_s.**AddCircle**($O_{p_{new}}$, S);
21: /* Edges connecting from p_{parent} to p_{new}. */
22: $\mathbb{E}_i$ = **Edge**(p_{parent}, p_{new});
23: T_s.**AddEdges**($\mathbb{E}_i$,$\mathbb{E}$);
24: /*Expand T_g utilizing basic RRT-connect mode.*/
25: T_g = **BidirectionExpansion**(p_{new}, T_s, T_g);
26: /*Rewire parent of nodes in P_{near} by Eq.3.11.*/
27: T_s = **Rewire**(p_{new}, T_s, T_g);
28: **Swap**(T_s, T_g);
29: **else**
30: Goto step 3;
31: **end if**
32: **end while**

In this context, $\mathbb{P}$ represents the probability, $L(\cdot)$ denotes the path length function while Π^* refers to the optimal solution.

Proof. A path that avoids collisions, represented as $\Pi : [0,1] \to \mathcal{C}_f$, is hereby defined. The trajectory originates from the starting area $\Pi(0) = p_s$ and terminates in the target area $\Pi(1) \in p_g$ while ensuring that $\Pi(\tau) \in \mathcal{C}_f$ for all $\tau \in [0,1]$. The length of the viable path can be formulated as follows:

$$L(\Pi) \geq L(\Pi^*)$$

The circle-RRT-connect* method, analogous to RRT* algorithm, utilizes critical strategies wherein the trees T_s and T_g update and optimize both the parent nodes of newly added nodes and neighbouring nodes in a designated region during the expansion procedure. The following condition is met by the $(n+1)$th iteration, as implied:

$$L(\Pi)_{n+1} \leq L(\Pi)_n$$

As the quantity of sampling points increase and $n \to \infty$, the likelihood that $L(\Pi)$ converges to $L(\Pi^*)$ approaches one:

$$\mathbb{P}(\{\lim_{n\to\infty} L(\Pi) = L(\Pi^*)\}) = 1.$$

Hence, the asymptotic optimality of the circle-RRT-connect* method is demonstrated. (Note that the radius adaptation in circle tree expansion does not impede the infinite sampling assumption.) □

Property 3.2. *Regarding path planning for an ASV navigating from the starting point p_s to the target point p_g, the circle-RRT-connect* algorithm exhibits a time complexity of $O(n \log n)$ and a space complexity of $O(n)$, enabling the effective recognition of a collision-free optimal path from the starting point to the intended point.*

Proof. The circle-RRT-connect* algorithm is developed based on the essential concepts underlying the RRT* technique. Key aspects of this method include generating random samples, identifying the closest node, selecting a parent, expanding T_s and T_g, and updating connections through rewiring. Assuming n represents the overall count of samples and v denotes the quantity of environmental obstacles. In each algorithm iteration, Performing the sampling process is a straightforward task, requiring constant time $O(1)$ to complete. The time complexity for searching the k-nearest neighbours is $O(\log n)$. The extension process of tree T_s exhibits a time complexity of $O(\log v)$, while the extension process of tree T_g is constrained to a time cost of $O(\log n)$. Given that the number of nodes within the circular area is limited relative to the total number of samples, the processes of choosing a parent and performing rewiring can each be regarded as having constant computational complexity, namely $O(1)$. The computational time required for the path planning process in circle-RRT-connect* can be represented as $O(n(1 + \log n + \log v + \log n + 1 + 1)) \approx O(n \log n)$. As a result, the total computational complexity of the circle-RRT-connect* approach is typically constrained by $O(n \log n)$. Because the number of sampling circles remains constant at n, the overall memory usage for the circle-RRT-connect* algorithm depends on the total storage required to keep all nodes and edges. This results in

$$|V| + |E| = n + (n-1) = 2n - 1,$$

leading to a space complexity for the circle-RRT-connect* method that can be briefly represented as $O(n)$. This completes the proof. □

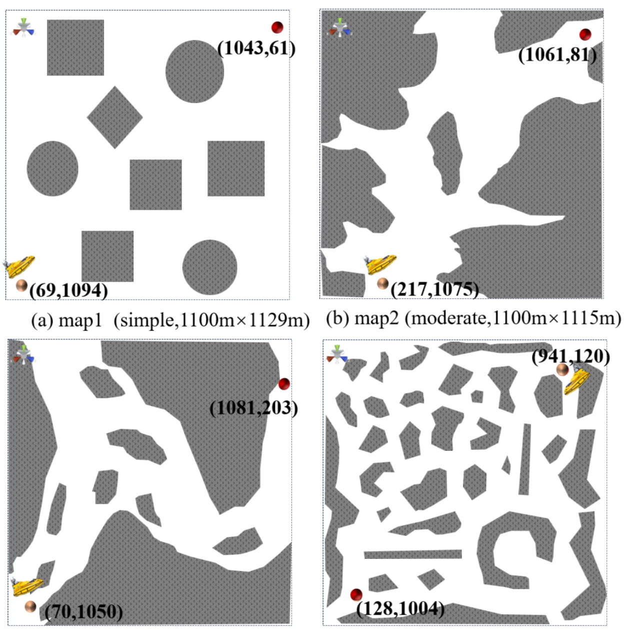
Kuffner et al. [12] established that the time complexity of RRT-connect is $O(n \log n)$. According to our mathematical analysis, the time complexity of the circle-RRT-connect* method is comparable to the standard RRT-connect. Nonetheless, circle-RRT-connect* provides a unique benefit by striking a harmonious balance between time effectiveness and optimal pathfinding. Comprehensive experimental results that support these findings are illustrated in Section 4.

4. Simulation and analysis

This section reports the experimental results, highlighting the effectiveness and energy-saving characteristics of our planning approach across different situations. All simulation experiments were implemented in Microsoft Visual C++ 2022, with no dependence on third-party numerical libraries. To maintain uniformity and repeatability, every test was performed on a computer equipped with an Intel i5-13500HX processor running at 2.50 GHz and 16 GB of RAM.

4.1. Performance

In order to showcase the performance of our proposed technique, we utilized four standard test environments, illustrated in Fig. 8. The test environments selected for planning display varying degrees of complexity, stemming from varied ocean current settings and obstacle arrangements, wherein certain scenarios pose greater challenges than others. For each test scenario, we compared the effectiveness of our method with the relevant methods described below.



(a) map1 (simple,1100m×1129m) (b) map2 (moderate,1100m×1115m)

(c) map3 (high,1100m×1115m) (d) map4 (very high,1100m×1110m)

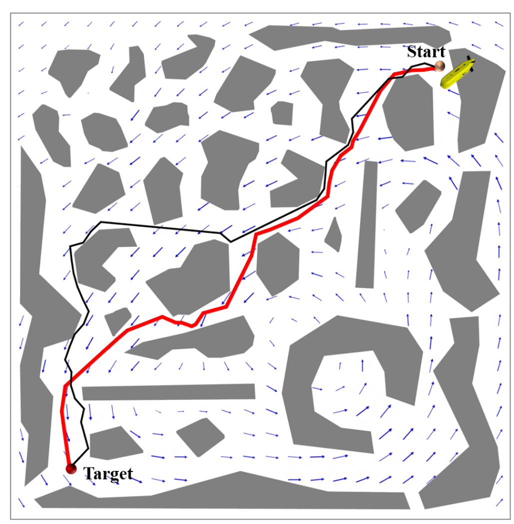
Figure 8: To showcase the efficacy of the suggested approach, four maps featuring varying degrees of complexity are presented.

- Ref. [11] introduced a cost model that measures how ocean currents affect energy usage and incorporates this into the D* path-planning technique. This method allows autonomous vehicles to utilize ocean currents efficiently, leading to a decrease in energy consumption.

- Ref. [47] suggested a fusion of heuristic algorithms that integrates genetic algorithms with ant colony optimization to improve path planning for autonomous vehicles, considering the influence of ocean currents.

- Ref. [12] enhances energy-efficient path planning for ASVs by integrating ocean current information and utilizing an energy-based path definition as proposed in the study.

Considering the intrinsic randomness inherent in RRT-based methods, We conducted 30 tests for each benchmark to ensure reliable and consistent results. Our analysis emphasizes three core metrics: energy expenditure considering ocean currents, the length of the path, and the required time. For each benchmark, these metrics are assessed by comparing the corresponding methods, as detailed in Table 1. As evidenced by the experimental findings obtained, our approach shows superior performance in comparison to other elegant methods. For example, we analyze the process of path planning as it occurs in the fourth scenario depicted in Fig. 9. The scenario features simulated ocean currents formed by the combination of four overlapping vortices, which are specified in Eq. 2.1. To achieve a trajectory measuring 1432m in length, our approach consumed 306.24J energy, while maintaining an average execution time of merely 0.184 seconds. In contrast, the related and elegant method proposed by Wen et al. [47] requires 337.32J energy and 2.140 seconds, while resulting in a path length of 1581m. When contrasted with the method proposed by Wen et al. [47], our method operates with greater efficiency, leading to notable decreases in energy consumption and a shorter path.

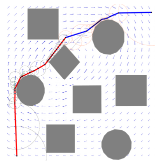
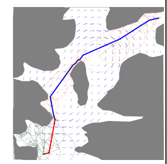
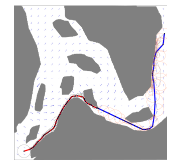


	Energy(J)	Time (s)	Length(m)
Ours	**306.24**	**0.184**	**1432**
Wen et al.[46]	337.32	2.140	1581

Figure 9: The visualization depicts path planning results generated by our method (in red) and the elegant approach (in black) proposed by Wen et al. [47]. The current speed ranges from 0 to 0.5 m/s.

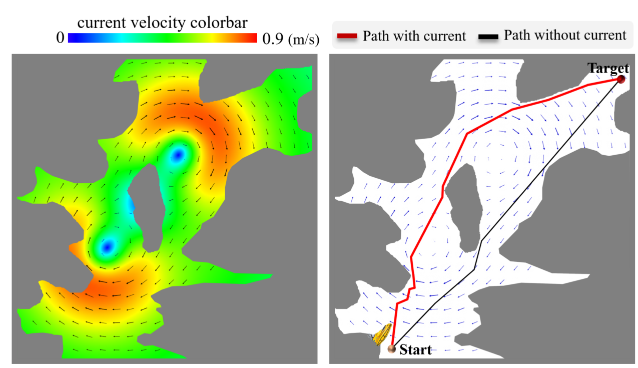
It is worth noting that in energy-efficient path planning for ASVs, the key factors influencing energy consumption include ocean currents, path quality and environment adaptivity. We conduct pertinent experiments of three critical indicators to highlight our method's advantages.

Table 1
A comparative evaluation of four interconnected methods conducted across different scenarios.

Map / Method	Assessment				
Ours	Energy cost(J)	**2121.24**	**322.45**	**582.71**	**305.26**
	Average Time (ms)	**10.63**	**38.93**	**75.61**	**179.86**
	Average Length(m)	**1715**	**1545**	**1823**	**1431**
Ref. [11]	Energy cost(J)	2415.70	389.22	731.93	375.21
	Average Time (ms)	101.73	508.02	943.28	1611.30
	Average Length(m)	1927	1801	1995	1664
Ref. [47]	Energy cost(J)	2201.91	344.60	617.31	324.99
	Average Time (ms)	147.11	785.94	1482.10	2572.91
	Average Length(m)	1793	1642	1881	1504
Ref. [12]	Energy cost(J)	2357.33	376.97	703.37	341.51
	Average Time (ms)	22.90	159.66	283.30	517.03
	Average Length(m)	1846	1745	1917	1569

Ocean currents. ASV path planning is greatly influenced by ocean currents. Overlooking the constraints imposed by these currents can lead to considerable energy losses, particularly in situations where the ASV must travel in opposition to the current. In order to improve the energy efficiency of ASVs that function under constrained energy supplies, we adjust the weighting factor for path length throughout the RRT tree growth process and employ a path formulation based on energy consumption, taking into account the influence of ocean currents. This approach reformulates the circle-RRT-connect* method to enhance the search for energy-efficient paths. We evaluated the outcomes of path planning using the circle-RRT-connect* technique, examining scenarios that either included or ignored the effects of ocean currents, as shown in Fig. 10. The results of the visualization distinctly reveal that considering ocean currents during route planning enables more effective ASV path design. By taking this factor into account, energy consumption is reduced when moving against the current, and there is also an increased opportunity to utilize the thrust generated by the currents themselves, thereby promoting energy efficiency.

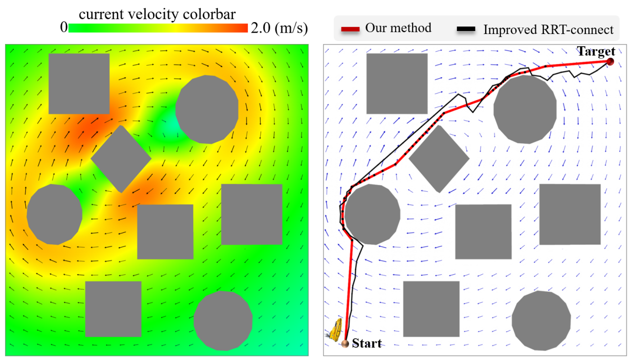
Path quality. For ASV path navigation, shorter paths under identical environmental conditions result in significantly reduced energy consumption during task execution. Path quality serves as a critical metric for evaluating the effectiveness of path-planning strategies. In this study, by capitalizing on the strengths of asymptotic optimality inherent in RRT*, we present a novel strategy to enhance path quality. To validate the optimality of our algorithm, we improve the basic RRT-connect method by incorporating the energy-based path framework combined with ocean currents. As shown in Fig. 11, although the improved RRT-connect accounts for the impact of ocean currents, it requires more energy due to the absence of a path rewiring process based on energy-conserving operations implemented in our proposed method.



(a) simulated current velocity field (b) path planning results

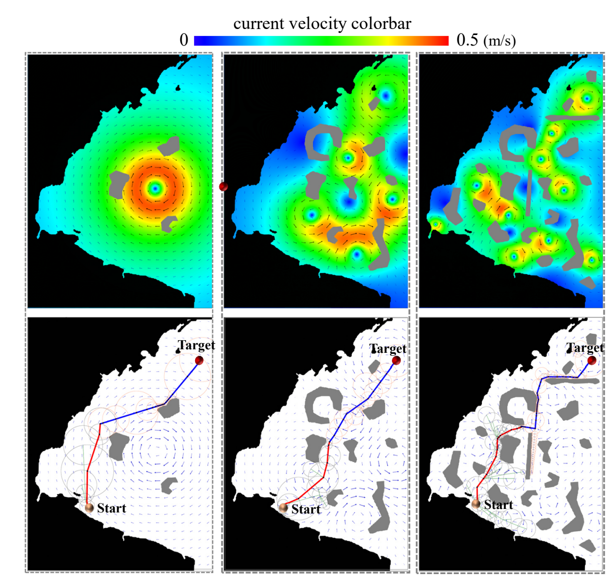
Figure 10: Path planning results utilizing the circle-RRT-connect* method both with and without considering ocean currents. (a) The simulated current is derived from two superimposed Lamb vortexes. (b) The circle-RRT-connect* method utilizes ocean currents to generate an energy-efficient path (red), requiring only 332.15J energy. In contrast, the path (black) obtained by the circle-RRT-connect* method without accounting for currents consumes 435.54J energy.

Environment adaptivity. The algorithm we propose demonstrates strong adaptability when faced with complicated environments that feature both complex obstacle arrangements and changing ocean currents, as shown in Fig. 12. This strength enables efficient navigation through demanding situations, maintaining both energy-saving path selection and stable, responsive operation. Its built-in flexibility allows for real-time adjustment to shifting currents and the emergence of obstacles, making it particularly well-suited for diverse marine applications.



(a) simulated current velocity field (b) path planning results

Figure 11: Path planning results utilizing the improved RRT-connect algorithm and our method in the context of the current velocity field. (a) The simulated current is derived from three superimposed Lamb vortexes. (b) The red trajectory, created using our approach, measures 1721m in length and requires 2257.4J energy. In contrast, the enhanced RRT-connect algorithm generates a black route that is longer, spanning 1873m in length, and demands a much greater energy expenditure of 2411.7J.

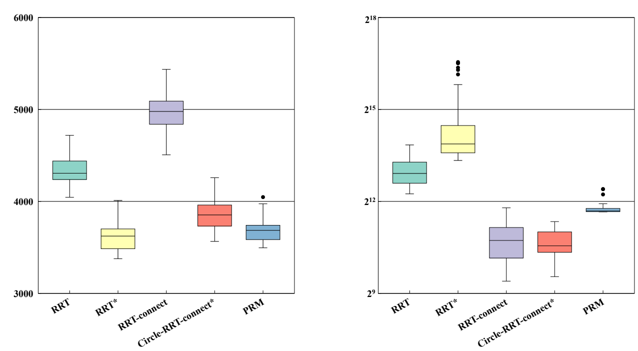


(a) Energy: 265.1J Time: 4.43ms (b) Energy: 154.1J Time: 6.86ms (c) Energy: 226.5J Time: 13.6ms

Figure 12: Path planning through complex environments involves intricate obstacle arrangements and changing ocean currents. The research covered three different scenarios: (a) basic obstacle layouts featuring one vortex, (b) moderately complex obstacle setups with five overlapping vortexes, and (c) highly intricate obstacle designs incorporating ten overlapping vortexes.

4.2. Circle-RRT-connect* evaluation

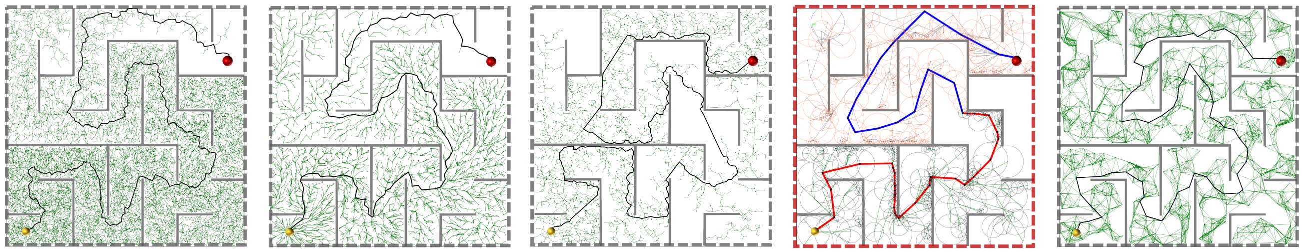
As previously noted, optimizing navigation paths under equivalent environmental conditions significantly reduces energy consumption during ASV task execution. Path length and efficiency are critical considerations for assessing the performance of path-planning strategies. For efficient path generation within the current field, we present an enhanced approach based on RRT-connect, which we call circle-RRT-connect*. The process of circle-RRT-connect* combines adaptive circle sampling with the standard RRT* rewiring technique, overcoming the limitations of conventional RRT-based methods and leading to enhanced time efficiency and optimized path lengths. We present a comparative analysis between the standard RRT algorithm [24], its derivatives including RRT* [13], RRT-connect [12], and our newly introduced circle-RRT-connect* approach. We also perform a comparative analysis with the standard PRM method [48], which is a well-established approach within the class of sampling-based algorithms. Figure 13 displays the statistical results related to path planning, employing the complex maze shown in Figure 14. The evaluation encompassed 30 trials for each method under comparison, documenting both the path length and time expended. The clear statistical outcomes show that, in comparison to other similar algorithms, the circle-RRT-connect* method efficiently yields paths that are close to optimal. This unique characteristic elevates the attractiveness of our introduced algorithm in path planning scenarios.



(a) Path length (b) Time cost

Figure 13: The statistical analysis summarizes the results obtained from path planning experiments conducted using the intricate maze shown in Fig. 14. Thirty trials were performed for each of the following methods: the basic PRM approach [48], as well as four RRT-based algorithms—RRT [24], RRT [13], RRT-connect [12], and circle-RRT-connect. For every trial, both the resulting path length and the computation time were meticulously recorded.

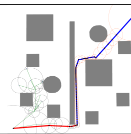
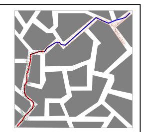
In particular, when compared with the standard RRT, the circle-RRT-connect* algorithm yields a path that demonstrates significant progress, most notably by shortening the route and increasing efficiency in terms of time. As illustrated in Fig. 14(a) and (d), this benefit becomes even more pronounced in intricate maps. Furthermore, while RRT* boasts the benefit of path optimality, it experiences sluggish convergence speeds. In contrast, circle-RRT-connect* excels in time efficiency, though it tends to generate slightly longer paths. By combining the optimal path planning capability of RRT* with the rapid exploration efficiency of RRT-connect, the proposed circle-RRT-connect* algorithm is able to generate paths that are close to optimal, demon-



(a) RRT (b) RRT* (c) RRT-connect (d) circle-RRT-connect* (e) PRM

Figure 14: Results obtained by the basic PRM method [48] and four RRT-based methods, including RRT [24], RRT* [13], RRT-connect [12] and circle-RRT-connect*. The start and goal points are represented by the yellow and red balls.

Table 2
We assess the performance of five advanced methods by analyzing their statistical results over various map scenarios.

Method \ Map	Assessment		
Ours	Time (ms)	**160.7**	**314.1**
	Length(m)	1463	1681
Standard RRT [24]	Time (ms)	1241.6	3254.2
	Length(m)	1822	1773
RRT* [13]	Time (ms)	4243.9	7730.6
	Length(m)	**1439**	**1673**
RRT-connect [12]	Time (ms)	245.1	449.8
	Length(m)	1833	1804
PRM [48]	Time (ms)	1232.4	2856.5
	Length(m)	1628	1730
DRL method [49]	Time (ms)	1765.5	4305.6
	Length(m)	1447	1677

strating excellent overall performance. As an illustration, in the intricate maze depicted in Fig. 14, the circle-RRT-connect* approach was able to quickly find a concise path, with an average length of $3874m$, and completed the planning process in approximately 1.31 seconds on average. Comparatively, while the RRT* algorithm resulted in a path of $3618m$ in length, the completion of the same task required a considerably longer duration of 41.28 seconds. The RRT-connect method produced a path measuring $4976m$ in length and completed the task in 1.78 seconds. These results unequivocally show that our proposed circle-RRT-connect* algorithm effectively combines the key features of both RRT* and RRT-connect, thus achieving a perfect balance between optimal pathfinding and time efficiency.

To offer a more thorough comprehension of the performance capabilities of circle-RRT-connect*, we performed a comparative analysis against several representative RRT-based approaches and the basic PRM method [48]. Two distinct benchmark scenarios were utilized in our experiments, with each scenario subjected to 30 independent trials. For every method evaluated, the mean values of computation time and path length were statistically analyzed. A complete summary of the findings is available in Table 2.

Table 3
Relationship between connection parameter ϵ, time and path length.

ϵ	Figure 8(a) Time(ms)	Figure 8(a) Path length(m)	Figure 8(d) Time(ms)	Figure 8(d) Path length(m)
5	22.14	1587	211.30	1321
20	9.23	1634	143.51	1377
40	7.03	1701	115.44	1424
80	5.82	1783	102.71	1489
120	4.57	1811	92.42	1504

4.3. Parameter sensitivity and tuning

The algorithm comprises multiple parameters whose tuning strategies are explicitly specified. The subsection quantify their influence on path length and computational time and present our recommended settings.

Connection parameter. In the circle-RRT-connect* algorithm, the search termination criterion is as follows: The circle-tree extension continues until the goal tree T_g is connected to the start tree T_s, satisfying $\|p_{new} - p_g\| \leq \epsilon$, where ϵ is a predefined connection threshold. Generally, a smaller ϵ results in shorter computed paths but significantly increases computation time. Conversely, a larger ϵ leads to longer paths but reduces computation time. This trade-off is demonstrated in Table 3, which shows experimental results across various map sizes. In this work, we selected an empirical value of $\epsilon = 10/(L_{map\_s})$ to balance path length and computation time, where L_{map_s} is the max value (height or width) of the map size. For tuning in different setups: In dense obstacle environments, reduce ϵ for precision; in open spaces, increase it for efficiency.

 # Radius $\mathcal{R}$ in* ChooseParent *for rewiring operation. In the *ChooseCircleParent* procedure of circle-RRT-connect*, a subset P_{near} is derived from the existing point set P . This subset is comprised of points within a hypersphere of radius $\mathcal{R}$, centered around p_{new}. The process identifies the optimal node p_{parent} from P_{near} to replace the candidate parent p_{near} of p_{new} (as illustrated in Figure 6(b)). The size of radius $\mathcal{R}$ is crucial for selecting an optimal parent node. A larger $\mathcal{R}$ expands the search range, increasing the probability of finding a better parent but consuming more time. This is quanti-

Table 4
Relationship between connection parameter $\mathcal{R}$, time and path length.

$\mathcal{R}$	Figure 8(c)		Figure 8(d)	
	Time(ms)	Length(m)	Time(ms)	Length(m)
5	41.86	1861	129.21	1522
10	69.44	1791	152.33	1409
20	104.33	1643	233.21	1311
40	167.90	1547	342.54	1255

fied in Table 4. In this work, we chose an empirical value of $\mathcal{R} = 3 * r_{p_{new}}$ (corresponding to circle radius of p_{new}) as the optimal radius. For tuning strategies: In complex setups with many obstacles, increase for better optimization; in simpler setups, decrease it to prioritize speed.

Drag Coefficient ***c.*** In Equation 3.5, the drag coefficient c represents the resistance factor in fluid dynamics, particularly for Autonomous Surface Vehicles (ASVs) moving through water. It is a dimensionless parameter that quantifies the drag influenced by the object's shape, surface roughness, fluid density, and velocity. For ASVs or similar watercraft, c typically ranges from 0.1 to 1.0. In this work, we selected a classical value of $c = 0.5$ for ASVs. For varying operational setups: In high-speed or turbulent water conditions, tune c upwards (e.g., 0.7–1.0); in calm environments, use lower values (e.g., 0.3–0.5) for efficiency.

4.4. Quantitative comparison

In this part, we carry out a comparison between our method and three other elegant methods in the field that resemble ours, emphasizing the unique benefits of our method.

4.4.1. Comparison to Ref. [11]

Sun et al. enhanced the conventional D* algorithm by incorporating additional factors, such as obstacle penalties and steering angle costs, into the objective function, thus improving navigational safety. Recognizing that ocean currents significantly impact ASV energy usage, they developed a cost model to integrate this factor into the D* path planning framework. This modification allows autonomous vehicles to utilize ocean currents, leading to reduced energy consumption. Graph-based techniques represent the environment as a grid and construct a network to identify viable paths. As a result, the initial setup stage often demands considerable computational resources. Employing graphs with finer resolution can yield more precise route exploration and shorter trajectories, yet this comes at the expense of increased computation. Therefore, these methods usually need to strike a compromise between precision and performance. By contrast, our approach utilizes circular sampling alongside rewiring mechanisms, which generates shorter routes and avoids the above-mentioned trade-off. For comprehensive performance analysis, refer to Table 1 and Figure 1.

4.4.2. Comparison to Ref. [47]

This work proposed an integrated approach that merges heuristic algorithms, enhancing the genetic algorithm by embedding ant colony optimization and simulated annealing methods. In addition, both an ocean current model and a kinematic framework were developed to address path-planning issues influenced by ocean dynamics. To improve convergence speed and expand the exploration range of the algorithm, several measures were adopted, including multiple crossover attempts, path self-smoothing processes, and dynamically adjusting the probabilities of genetic operations. Nevertheless, as demonstrated in the experimental findings (see Table 1), when applied to large-scale environments, this technique incurs significant computational overhead for fitness evaluation. Furthermore, its performance is highly dependent on parameter choices, such as mutation and crossover rates, necessitating careful tuning for optimal results. By comparison, our method offers a more straightforward and computationally efficient solution.

4.4.3. Comparison to Ref. [12]

Kuffner et al. developed the RRT-connect algorithm, an improved variant of RRT aimed at achieving more effective path planning. This approach constructs two trees in parallel: one starts from the initial location, while the other grows from the destination. The algorithm alternates between expanding each tree in the direction of the other, with the goal of eventually linking them. Once the trees meet, a feasible route is established. This dual-tree strategy significantly shortens search times compared to the original single-tree RRT. By integrating an energy-focused path framework that incorporates the influence of ocean currents, as described in the study, the RRT-connect algorithm demonstrates notable capability in energy-conscious path planning. Nevertheless, it should be noted that even with the inclusion of ocean current effects, the enhanced RRT-connect still consumes more energy. This is primarily because it lacks a path rewiring mechanism that specifically targets energy reduction, a feature present in our method. A comprehensive comparison of the results can be found in Fig. 11 and Table 1.

5. Conclusion

In this research, we introduce a novel approach that effectively tackles the demanding challenge of sustainable navigation for ASVs operating in intricate ocean current conditions. When compared with other prominent techniques, our algorithm demonstrates several clear advantages. The strengths of our method are outlined below: (1) We propose a new energy-conscious path planning strategy for ASVs, which simultaneously accounts for multiple elements,including obstacle avoidance, distance evaluation, and dynamic ocean currents, to achieve optimal route generation. (2) By integrating the asymptotic optimality characteristic of RRT* with the fast convergence capability of RRT-connect, we have developed an enhanced algorithm named circle-RRT-connect*. This new version achieves an effective compromise between route quality and computational speed. (3) The influence of ocean currents is directly embedded within the path planning framework, which improves the growth behavior of the RRT tree and utilizes current-driven propulsion to reduce energy consumption. After conducting a series of thorough simulations, we have demonstrated the superior performance of our algorithm in ensuring sustainable navigation in complex oceanic environments, and we have benchmarked its results against those of other state-of-the-art methods.

A key assumption in this work is the availability of accurate environmental perception, which simplifies the evaluation of our core contributions but may not hold in real-world scenarios where sensors are prone to noise, occlusions, or limitations. This could lead to suboptimal paths or increased energy consumption if uncertainties are not accounted for. To mitigate the limitations, future work could integrate probabilistic models of sensing uncertainty (e.g., Bayesian filtering for ocean current estimation) directly into the circle-based sampling and rewiring processes, enabling more robust planning. While our model effectively captures the free propagation of Lamb vortices, it does not incorporate interactions with obstacles, such as walls or solid bodies, which could introduce effects like vortex reflection, dissipation, or induced secondary vortices in real-world applications. Future extensions could integrate boundary element methods or immersed boundary techniques to model these interactions more accurately.

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